

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457063.89283 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.057
Transit depth (Rp/Rs)^2	2.05 +/- 1.64 [%]
Orbital Inclination (inc)	87.12 +/- 2.82
Airmass coefficient 1 (a1)	0.18 +/- 0.11
Airmass coefficient 2 (a2)	1.02 +/- 0.66
Scatter in the residuals of the lightcurve fit is	64.42 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	3.0
Transit Duration (day)	0.0584 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.143 ± 0.057 vs 0.1594 (deviation 0.22σ)
Duration fit vs published	0.0584 d vs 0.0512 d (deviation 1.14σ)
Mid-time O-C	-2.1 min
Depth SNR	2.0
Residual scatter	64.42 %

Light curve

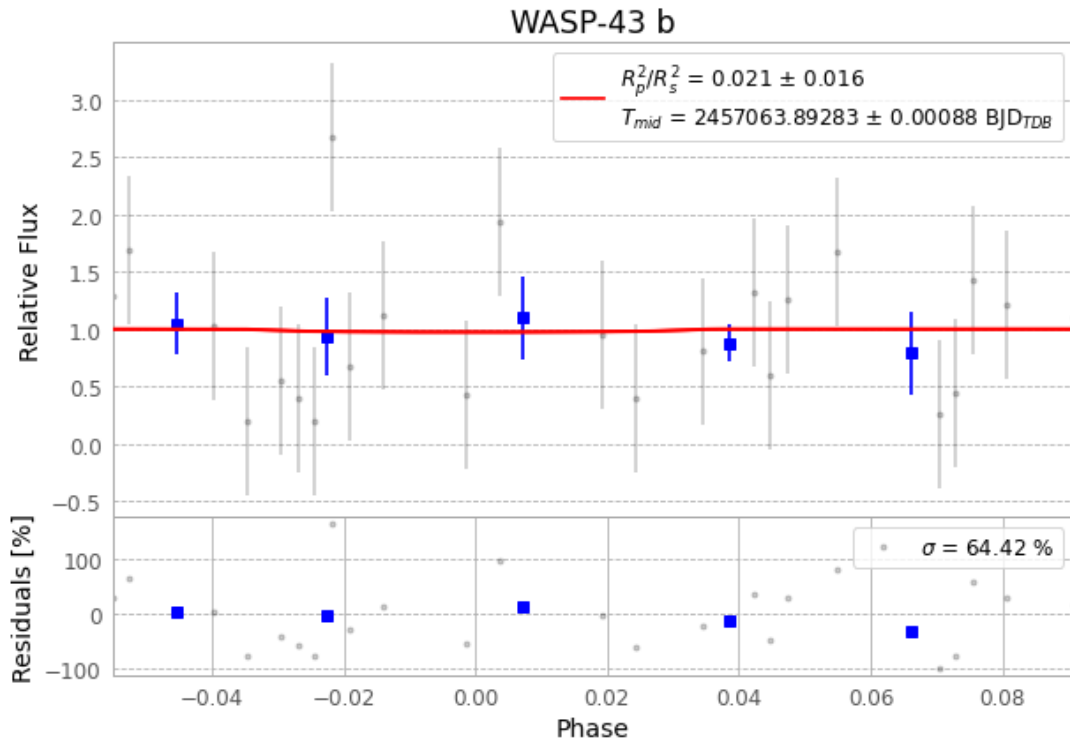


Figure 1 — FinalLightCurve_WASP-43 b_2015-02-10.png (SHA-256 4ec40f8c01b67654...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [401, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7acf7d13c26a05c3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-43150210075712.FITS → WASP-43150210110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150210.log (80561c5102b0...), FinalLightCurve_WASP-43 b_2015-02-10.png (4ec40f8c01b6...), FinalLightCurve_WASP-43 b_2015-02-10.csv (e39e684f4913...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:49Z.